

Technology Stack

Script-Controlled ACL — Restrict Record Access Based on Field Value

1. Platform & Data Layer

Component	Technology	Purpose
ServiceNow Platform	Now Platform (Global scope)	Core PaaS hosting the application, tables, roles, and ACL engine
Custom Table	u_institution_details	Stores Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, Description
Field Types	Auto Number, Reference (User), Choice, String, Multi-line String	Define how each column of the table stores and validates data

2. Security & Access Control Layer

Component	Technology	Purpose
Access Control	Script-controlled ACLs (record-level)	Restricts Read / Create / Write / Delete on u_institution_details
Scripting Engine	Server-side JavaScript (Rhino/GlideScript, ES12 optional)	Runs ACL condition scripts such as gs.hasRole() checks
GlideSystem API	gs.hasRole('admin'), gs.hasRole('bb1'), etc.	Server-side API used to evaluate the current user's roles inside ACL scripts
Roles	admin, bb1 (Read), bb2 (Create), bb3 (Write), bb4 (Delete)	Custom + built-in roles required by each ACL operation
Data Condition	Branch is EEE	Row-level filter combined with role check on the Read ACL
Elevated Privilege	security admin (Elevate Role)	Temporary elevation required to create/modify ACL records

3. Presentation / UI Layer

Component	Technology	Purpose
List View	ServiceNow List UI	Displays filtered institution_details records based on ACL evaluation
Form View	ServiceNow Form UI	Used for record creation/editing, gated by Create and Write ACLs
Service Portal	Now Experience / Service Portal	Alternate front-end surface where the same ACL rules apply
Impersonation	User Administration — Impersonate User	Used to test and verify ACL behaviour as different personas (EEE User, Admin)

4. Administration & Supporting Tools

Component	Technology	Purpose
User & Role Management	User Administration module	Create test users (EEE User) and custom roles (bb1-bb4)
Application Navigator	System Definition, System Security modules	Used to reach Tables, Access Control (ACL), Roles, and Users screens
Update Set	ACL Project [Global]	Tracks and packages all table, role, and ACL configuration changes made during the lab

5. Summary

The solution is built entirely on the ServiceNow Now Platform, without any external systems or third-party integrations. Data lives in a custom table secured by row-level, script-controlled ACLs written in server-side JavaScript. Access decisions combine ServiceNow's native role framework (gs.hasRole()) with a field-value data condition, enforced consistently across list view, form view, and Service Portal.